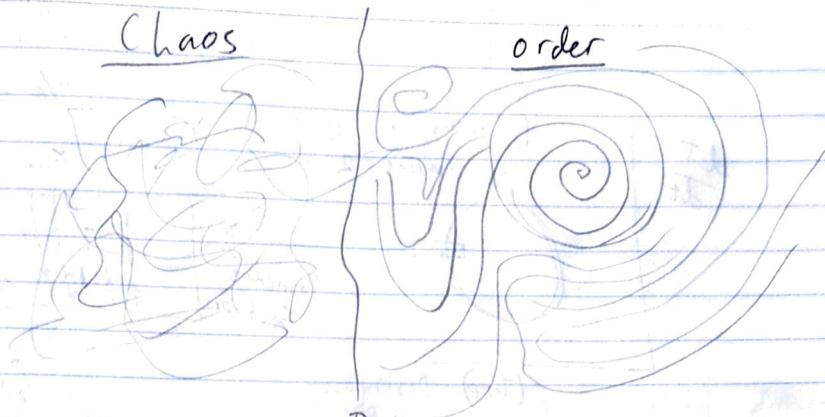


Chaos

May 6, 1976

Border of order



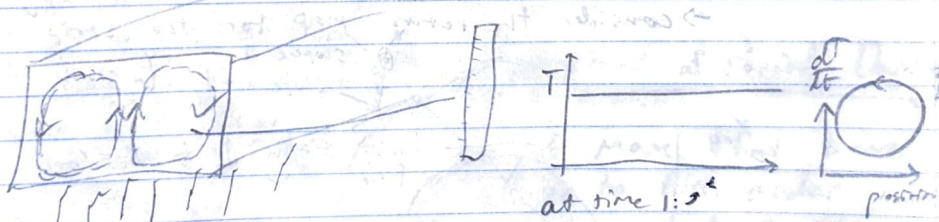
can we describe this transition?

→ The critical step is renormalization theory

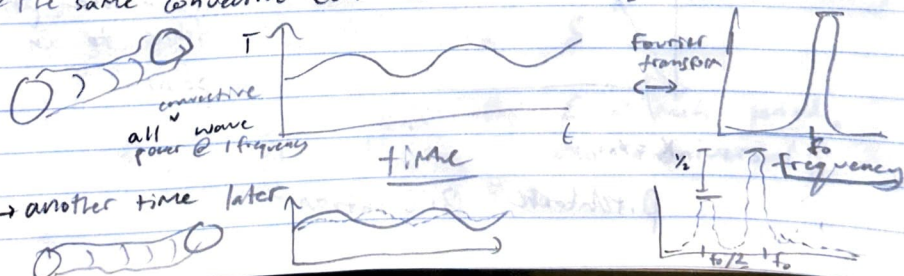
→ consider the transition to turbulence

- e.g., a heated pot of oil. no movement, then regular, then turbulent
- different characteristics for different settings

→ consider a reservoir heated from below

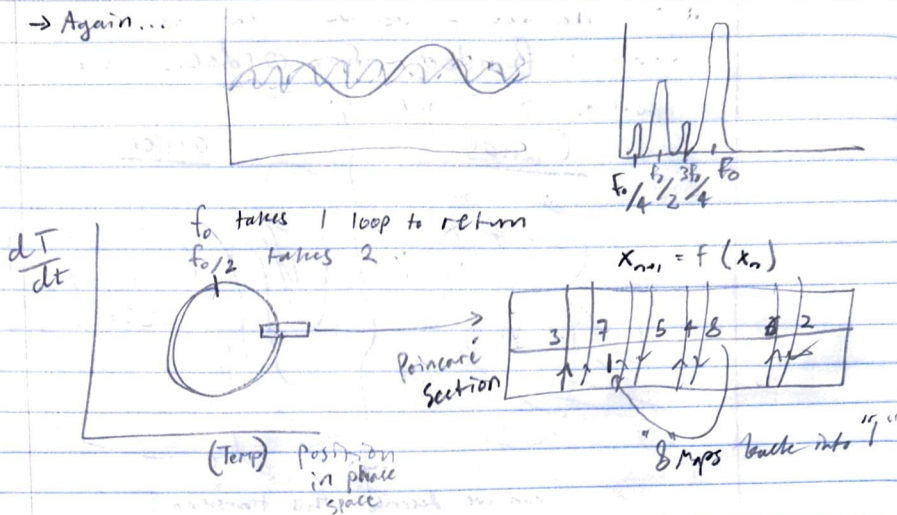


→ the same convective cell at a time T_2 later



→ another time later

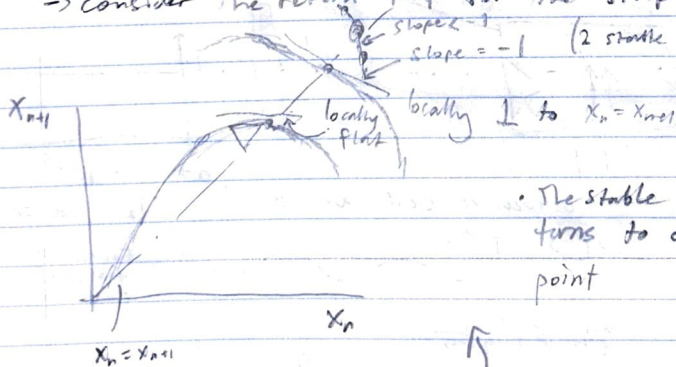
→ Again...



→ As the chaos develops ...

- mapping will grow to more and more values at rational frequencies
- the transition from regularly spaced frequencies to noise/mixing is evident audibly.
- "period-doubling" continues

→ consider the return map for the strip in $dT/dt/T$ ^{Poincaré section}



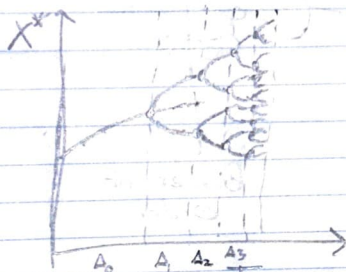
- The stable point $X_n = X_{n+1}$ turns to an unstable point

→ If the double-period returns to the same state after 2 iterations, consider the renormalized mapping $x_{n+2} = g(x_n)$:



→ consider prime map on next axis...

→ A consistent difference between splitting/period doubling



$$\frac{\Delta_n}{\Delta_{n+1}} \rightarrow \underline{\underline{\delta = 4.6692...}}$$

→ Feigenbaum computed this number w/ precision...!

$\delta \Leftrightarrow$ Universal Constant of Period Doubling

→ generic universality class of maps that have a quadratic local behavior to first order (almost all systems) admit period doubling at this exact ratio δ

→ Consider the ratio of scales ε at each period, also $\sim$ universal!

$$\frac{\varepsilon_n}{\varepsilon_{n+1}} \rightarrow \underline{\underline{-2.5029 \equiv \alpha}}$$

→ In general, the period doubling follows the generic limit of renormalized behavior such that the map is described by

$$\begin{aligned}g(x) &= -\alpha g(g(\frac{x}{\alpha})) \\ &= -\alpha g \circ g(\frac{x}{\alpha})\end{aligned}$$

→ Thus, the border of order:



Border of
Order

α, δ : Feigenbaum numbers

renormalization: $g(x) = -\alpha g \circ g(\frac{x}{\alpha})$